CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Dividing cells of a root tip are best seen under the microscope in the: 1

(a) growing region **(b)** root hair zone **(c)** root cap **(d)** mature root

2. A stem with a girth increasing yearly has active: 1

(a) root cap **(b)** apical meristem **(c)** vascular cambium **(d)** stomata

3. A plant is described as having vascular bundles scattered in the stem, parallel venation and fibrous roots. It is a: 1

(a) dicot **(b)** monocot **(c)** pteridophyte **(d)** gymnosperm

4. An egg is fertilised on the 14th day of the menstrual cycle. The next menstruation will be: 1

(a) on the 15th day **(b)** on the 28th day **(c)** absent **(d)** on the 21st day

5. Which of these processes removes nitrogen gas from the air for plants? 1

(a) burning **(b)** respiration

(c) transpiration **(d)** nitrogen fixation

6. An animal cell placed in pure water will: 1

(a) remain same **(b)** swell and may burst

(c) shrink **(d)** divide

7. A student sees dead cells with thick lignified walls in a section. The tissue is: 1

(a) parenchyma **(b)** collenchyma **(c)** sclerenchyma **(d)** phloem

8. An organism is multicellular, has a cell wall of chitin and feeds on dead matter. It is a: 1

(a) bacterium **(b)** plant **(c)** animal **(d)** fungus

9. A gardener wants a rose plant with flowers of two colours. The method is: 1

(a) seed sowing **(b)** grafting **(c)** spore formation **(d)** binary fission

10. Which of the following reduces your carbon footprint the most? 1

(a) using public transport and saving energy **(b)** burning waste

(c) using air conditioners all day **(d)** using plastic

11. A cell with 8 chromosomes divides by meiosis. Each daughter cell has: 1

(a) 4 **(b)** 2 **(c)** 8 **(d)** 16

12. A tissue of long cells with cilia on the surface is found in the: 1

(a) trachea **(b)** kidney **(c)** skin **(d)** stomach

13. An animal that lays eggs, has feathers and a beak is a: 1

(a) bird **(b)** mammal **(c)** reptile **(d)** amphibian

14. Papaya and cucumber flowers are: 1

(a) asexual **(b)** sterile **(c)** bisexual **(d)** unisexual

15. If the Earth received 1000 W per square metre and reflected 30 per cent, the radiation absorbed is: 1

(a) 970 W per square metre **(b)** 300 W per square metre

(c) 700 W per square metre **(d)** 1030 W per square metre

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A cell has 12 chromosomes. How many are present in the cells formed by mitosis and meiosis? 2

17. Why does cutting the bark in a ring (girdling) kill the tree after some time? 2

18. Why is a whale classified with mammals though it lives in water? 2

19. Why is a bud formed in Hydra considered a form of asexual reproduction? 2

20. Why do pulses (legumes) enrich the soil with nitrogen? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. An animal has a four-chambered heart, feathers and hollow bones. Identify the class and mention how the bones are adapted. 3

22. Compare the reproduction in hen and the reproduction in humans. 3

23. Explain how the human activities have disturbed the nitrogen cycle. 3

24. Explain why nerve cells are long and branched, using structure-function relation. 3

OR

Describe the relationship between tissues, organs and organ systems with the example of the stomach. 3

25. Describe the features of Platyhelminthes and the diseases caused by the parasitic forms. 3

OR

Explain the importance of vaccination and a healthy diet in pregnancy. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. The Sun's radiation of 1000 W/m squared falls on a surface of 2 square metres. Find the energy received in 1 hour and the energy absorbed by 5

(a) snow with albedo 0.85 and

(b) forest with albedo 0.1 in kWh.

OR

Explain how the cell is organised as a miniature factory, giving the roles of the nucleus, ribosomes, ER, Golgi, mitochondria, lysosomes and the membrane. 5

27. Describe the adaptations of the tissues in aquatic, desert and land plants with examples. 5

OR

Describe the adaptations of the birds for flight and of the fish for aquatic life with examples. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND